Trainings

Preparatory Steps

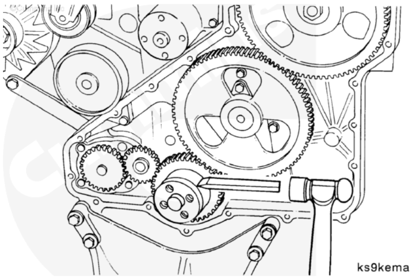
Remove the following:

- Drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html).
- Vibration damper. Refer to Procedure 001-052 (/qs3/pubsys2/xml/en/procedures/100/100-001-052.html).
- Front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/100/100-001-031.html).

Remove

⚠ CAUTION ⚠

Do not nick or gouge the crankshaft with the chisel. If the crankshaft is damaged, it must be replaced.



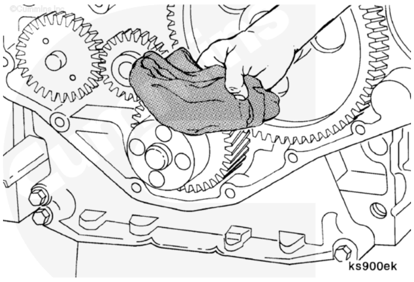
Note : Use a hammer and chisel that is only as wide as the wear sleeve.

Make one or two chisel marks across the wear sleeve. This will expand the wear sleeve, allowing the sleeve to be removed.

Clean and Inspect for Reuse

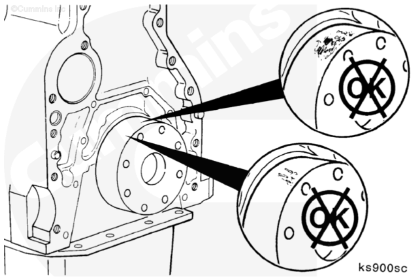
Use Scotch-Brite™ 7448, Part No. 3823258, to remove any rust or other deposits from the crankshaft flange.

Use a clean cloth to clean the crankshaft flange.



Inspect the seal contact area on the crankshaft for a wear groove. If the seal has worn a groove deep enough to be felt with a sharp object or fingernail, it will be necessary to install a wear sleeve to prevent an oil leak.

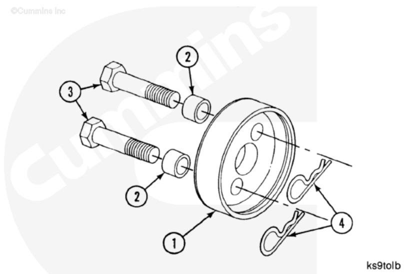
The oil seal used with the wear sleeve has a larger inside area than the standard seal. The two seals are **not** interchangeable.



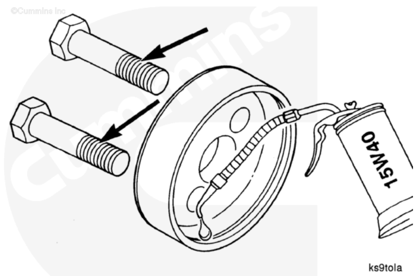
Install

Use the crankshaft front wear sleeve installation tool, Part No. 3823908, to install the wear sleeve to the correct position on the crankshaft. The tool kit consists of the following:

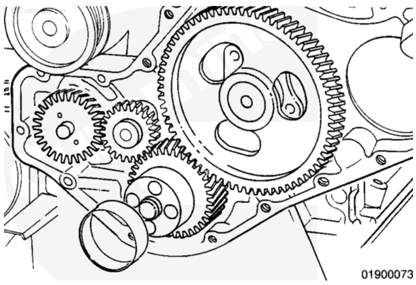
1. Driver - 1
2. Spacers - 2
3. Capscrews - M14x1.5x60 mm - 2
4. Hair Pin Cotter Pins - 2



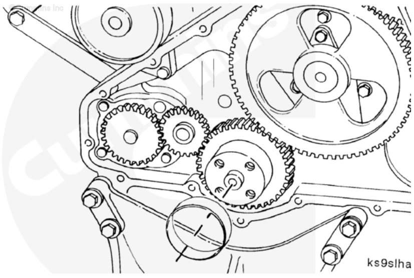
Apply a thin coat of clean engine oil to the inside of the driver and to the capscrew threads.



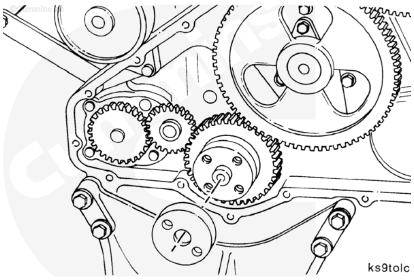
Apply a thin coat of clean engine oil to the crankshaft flange.



Position the chamfered end of the wear sleeve onto the end of the crankshaft.



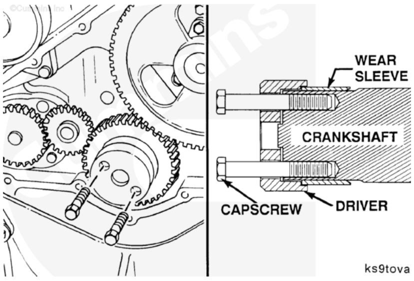
Position the counterbore end of the driver onto the wear sleeve.



Install two M14x1.5x60 mm capscrews (without spacers or hairpin cotter pins) through the driver and into the crankshaft capscrew holes.

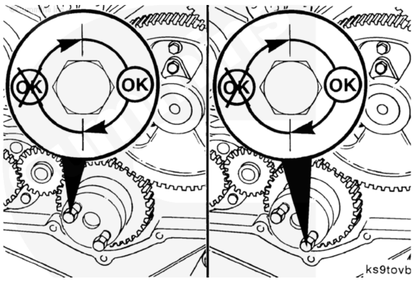
Align the wear sleeve and driver perpendicular with the crankshaft.

Tighten the capscrews “finger-tight.”



⚠ CAUTION ⚠

To prevent damage to the wear sleeve, do not exceed 1/2 of a revolution of each capscrew.

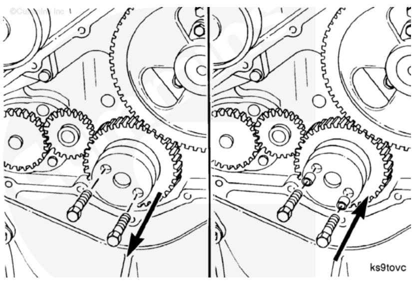


Alternately, tighten the capscrews until the sleeve is installed to a depth of approximately 16 mm [0.625 in].

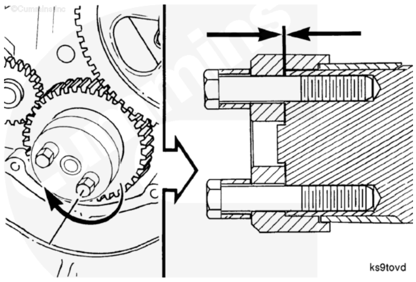
Torque Value: 20 n•m [15 ft-lb]

Remove the capscrews and install the spacer on each capscrew.

Install the two M14x1.5x60 mm capscrews.

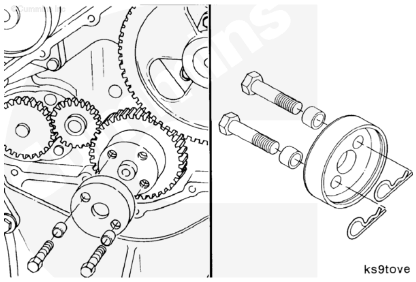


Continue to tighten the capscrews alternately until the bottom of the driver contacts the end of the crankshaft.



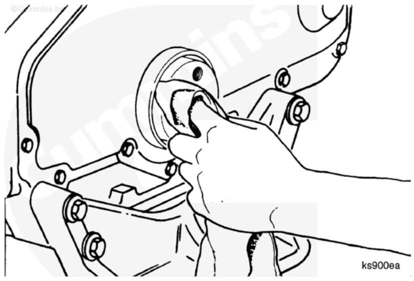
Remove the driver.

Note : Use the hairpin cotter pins to secure the capscrews and spacers to the tool during storage.



Clean the wear sleeve and crankshaft of any excess lubricants.

Note : The wear sleeve and oil seal mating surface must be clean and dry to allow proper oil sealing.



Finishing Steps

Install the following components:

- Front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/100/100-001-031.html).
- Vibration damper. Refer to Procedure 001-052 (/qs3/pubsys2/xml/en/procedures/100/100-001-052.html).
- Fan pulley. Refer to Procedure 008-039 (/qs3/pubsys2/xml/en/procedures/100/100-008-039.html).
- Drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html).

Operate the engine and check for leaks.

Last Modified: 20-Aug-2003